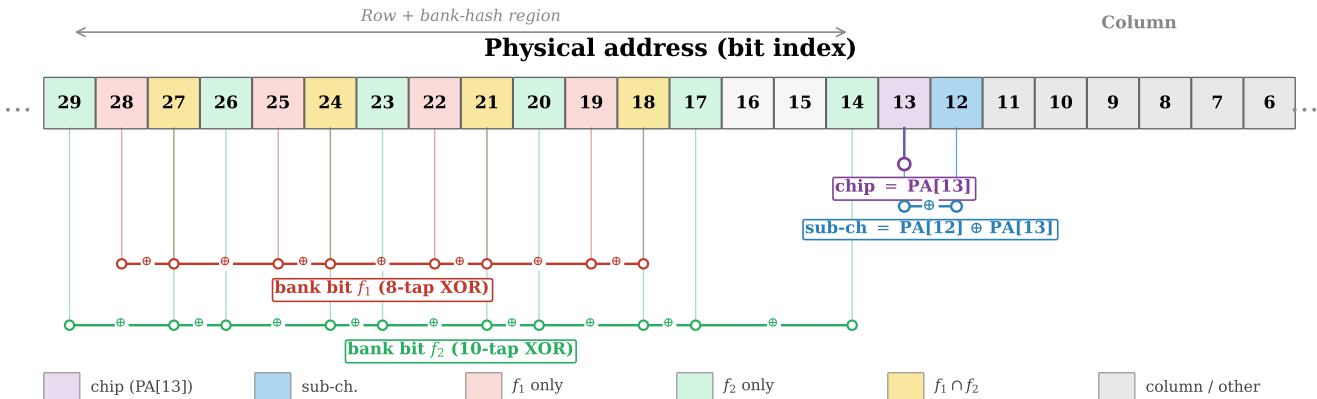


## (a) Recovered Tegra186 DRAM address mapping



## (b) Complete GF(2) linear mapping: $\text{PA} \mapsto (\text{chip}, \text{sub-ch.}, f_1, f_2)$

$$\begin{array}{c}
 \text{chip} \\
 \text{sub-ch.}
 \end{array}
 \begin{array}{c}
 \text{PA[12] PA[13]} \\
 \begin{bmatrix}
 0 & 1 \\
 1 & 1
 \end{bmatrix}
 \end{array}
 \begin{array}{c}
 \text{PA[12]} \\
 \text{PA[13]}
 \end{array}
 = \begin{array}{c}
 \text{chip} \\
 \text{sub-ch.}
 \end{array} \pmod{2}$$
  

$$\begin{array}{c}
 f_1 \\
 f_2
 \end{array}
 \begin{array}{c}
 \text{PA[14] PA[17] PA[18] PA[19] PA[20] PA[21] PA[22] PA[23] PA[24] PA[25] PA[26] PA[27] PA[28] PA[29]} \\
 \begin{bmatrix}
 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\
 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1
 \end{bmatrix}
 \end{array}
 \begin{array}{c}
 \text{PA[14]} \\
 \text{PA[17]} \\
 \vdots \\
 \text{PA[28]} \\
 \text{PA[29]}
 \end{array}
 = \begin{array}{c}
 f_1 \\
 f_2
 \end{array} \pmod{2}$$